

NORTH ADAMS, MA, USA

WMO: 725075

Lat: 42.697N

Lon: 73.170W

Elev: 195

StdP: 99.00

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 54768

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-16.1	-24.8	0.4	-17.6	-22.4	0.5	-14.8	11.7	-4.3	10.8	-4.1	2.7	270	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	30.5	22.0	29.0	21.1	27.7	20.5	23.4	28.2	22.6	27.1	21.8	25.9	2.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.1	17.2	25.9	21.2	16.2	25.0	20.3	15.3	24.2	70.6	28.1	67.3	26.8	64.2	25.9	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.4	7.5	-23.2	32.9	3.2	1.1	-25.5	33.6	-27.3	34.3	-29.1	34.9	-31.4	35.7		
(5)			-23.6	25.3	3.0	1.1	-25.8	26.1	-27.5	26.7	-29.2	27.3	-31.4	28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-4.9	-3.2	0.9	7.7	13.9	18.3	20.7	19.9	16.2	9.9	4.1	-1.1		
	DBStd	10.05	6.50	5.56	5.51	4.82	4.24	3.47	2.79	2.88	3.84	4.55	4.95	5.38		
	HDD10.0	1821	462	369	286	100	11	0	0	0	2	59	187	346		
	HDD18.3	3777	719	602	539	321	150	42	9	17	86	262	427	603		
	CDD10.0	1315	1	1	4	31	131	251	333	307	188	57	10	1		
	CDD18.3	229	0	0	0	2	11	43	84	66	22	2	0	0		
	CDH23.3	2106	0	0	6	43	181	424	754	506	178	13	0	0		
	CDH26.7	514	0	0	1	11	43	114	203	104	38	1	0	0		
(14) Wind	WSAvg	2.7	3.6	3.6	3.5	3.3	2.5	1.9	1.5	1.4	1.7	2.5	3.1	3.6		
(15) Precipitation	PrecAvg	1238	86	72	103	94	109	109	116	113	116	112	104	99		
	PrecMax	1555	144	186	166	160	202	193	234	242	191	310	156	171		
	PrecMin	915	45	41	36	31	27	16	49	45	49	31	63	36		
	PrecStd	171	29	31	31	33	43	45	49	55	46	64	28	36		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.5	15.4	23.2	28.2	30.1	32.1	32.4	31.3	30.2	25.0	20.8	16.0		
		MCWB	12.0	10.9	13.9	16.8	19.7	22.7	23.3	23.3	22.4	18.1	16.1	12.6		
	2%	DB	9.2	10.7	16.3	23.4	27.8	29.6	30.3	29.0	27.4	22.4	17.6	12.2		
		MCWB	6.5	6.7	10.3	13.8	18.5	21.0	22.3	21.8	20.7	17.0	13.3	9.9		
	5%	DB	6.2	7.0	12.4	20.1	25.3	27.7	28.9	27.7	25.4	20.1	15.0	8.9		
		MCWB	4.1	3.6	7.6	12.4	16.9	20.2	21.4	21.2	19.6	15.8	11.3	6.9		
	10%	DB	3.7	4.1	9.1	17.1	22.7	25.9	27.5	26.4	23.0	17.6	12.3	6.3		
		MCWB	1.8	1.6	5.3	10.5	15.7	19.0	20.7	20.2	18.3	13.8	9.4	3.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	12.1	14.4	18.5	21.3	24.0	24.7	24.5	23.3	20.3	16.9	13.6		
		MCDB	14.0	15.4	21.3	24.1	27.8	29.7	30.1	29.1	28.1	23.2	19.8	15.8		
	2%	WB	7.5	6.8	11.2	15.7	19.8	22.6	23.4	23.2	21.9	18.4	14.1	10.0		
		MCDB	8.7	9.6	15.1	21.8	25.6	27.7	28.3	27.4	26.2	21.3	16.8	11.9		
	5%	WB	4.2	3.9	8.0	13.6	18.5	21.6	22.6	22.3	20.6	16.3	11.9	7.2		
		MCDB	6.1	6.4	11.4	18.8	23.6	26.1	27.1	26.1	24.0	19.3	14.3	9.0		
	10%	WB	2.0	2.1	5.6	11.0	17.0	20.2	21.6	21.3	19.3	14.4	9.5	4.4		
		MCDB	3.6	4.2	8.9	15.8	21.5	24.2	25.9	24.7	22.3	17.1	12.1	6.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.9	10.2	12.0	12.4	11.8	11.8	11.6	11.9	10.8	9.4	8.0		
		MCDBR	10.7	13.1	15.4	18.0	16.7	14.7	13.4	12.7	14.0	14.4	13.4	11.0		
	5% WB	MCWBR	9.4	9.9	9.8	9.8	8.5	7.4	6.4	6.6	7.8	9.1	9.8	9.7		
		MCWBR	10.4	12.3	13.8	15.1	14.6	12.7	11.6	10.8	11.8	12.5	11.2	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.311	0.330	0.354	0.388	0.434	0.453	0.426	0.369	0.345	0.310	0.289		
	taud		2.380	2.326	2.383	2.402	2.342	2.236	2.194	2.253	2.396	2.474	2.522	2.480		
	Ebn at Noon		845	887	913	914	886	842	820	832	861	842	828	825		
	Edh at Noon		79	99	106	113	123	137	142	129	104	85	70	66		
(44) All-Sky Solar Radiation	RadAvg		1.62	2.50	3.56	4.53	5.13	5.55	5.78	5.17	4.20	2.64	1.80	1.29		
	RadStd		0.13	0.24	0.28	0.50	0.56	0.50	0.39	0.27	0.34	0.28	0.18	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+95	+44

Nomenclature: See separate page